

The Marine Environment

The Marine Environment and Marine Environment Protection

Exercise 1

The marine environment is a <**saltwater**> environment.

The marine environment means the natural and biological <**resources**>, comprising any marine ecosystem within the Exclusive Economic Zone's waters, <**seabed**> and subsoil including both the living and nonliving components of marine ecosystems.

The marine environment can also be described as the oceans, seas, estuaries, <**bays**> and other major water bodies, with the atmosphere and with the land seaward of the <**mean high water**> mark.

The marine environment means the physical, <**atmospheric**>, and biological components, conditions and factors which interactively determine the productivity, state, condition and quality of the marine ecosystem, including the <**waters of the high seas**>, the contiguous zone, transitional and intertidal areas, salt marshes and <**wetlands**> within the coastal zone and on the Outer <**Continental Shelf**>.

Marine environments include estuaries, coastal marine and <**nearshore**> zones and open-ocean-deep-sea regions

Exercise 2

All seafarers must be conscious of the pollution **threats** to **waterways and oceans** and the serious effects that may result.

MARPOL covers accidental and **operational** oil pollution as well as pollution by **chemicals**, goods in packaged form, **sewage**, garbage and **air pollution**.

The IMO's Intervention Convention affirms the right of coastal states to take measures on the **high seas** to prevent, **mitigate** or eliminate dangers to their coastlines from marine **casualties**.

The International Convention on Oil Pollution Preparedness, Response and Co-operation (OPRC) provides a global framework for international co-operation in **combating** major incidents or threats of marine pollution. A protocol to this convention covers marine pollution by **hazardous** and **noxious** substances (HNS).

The Marine Environment **Protection** Committee (MEPC) is IMO's senior technical body on marine pollution related matters. This Committee is aided in its work by a number of **Sub-Committees**.

Exercise 3

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MARPOL covers <**accidental**> and operational oil pollution as well as pollution by <**chemicals**>, goods in <**packaged**> form, sewage, <**garbage**> and air pollution.

The IMO's <**Intervention**> Convention affirms the right of <**coastal**> states to take measures on the <**high seas**> to prevent, mitigate or <**eliminate**> dangers to their <**coastlines**> from marine casualties.

The International Convention on Oil Pollution <**Preparedness**>, Response and <**Co-operation**> (OPRC) provides a global <**framework**> for international co-operation in <**combating**> major incidents or threats of marine pollution. A protocol to this convention covers marine pollution by hazardous and <**noxious**> substances (HNS).

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The Marine Environment and Man-made Pollution

Exercise 1

Marine pollution	occurs when harmful effects result from the entry into the ocean of chemicals, particles, waste, or the spread of invasive organisms.
Pollutants	are waste material that pollute air, water or soil.
Point source pollution	occurs when discharges, namely man-made contaminants, enter a water body at a single location.
(Surface) runoff	is the water flow which occurs when soil is saturated and excess water overflows into the land.
Something is 'toxic'	when it is chemically harmful or even deadly.

Plankton	are drifting organisms that inhabit the oceans or seas.
Benthos	are organisms which live on, in, or near the seabed.
Filter feeders or suspension feeders	are animals that feed by straining suspended matter and food particles from water.
Toxic chemicals are concentrated upward	when animals take in more toxics that they excrete.
Food chains or food networks	describe the eating relationships between living species within an ecosystem.

Estuaries	are semi-enclosed coastal bodies of water with one or more rivers or streams and with a free connection to the open sea.
The term 'anoxic'	refers to water that is depleted of dissolved oxygen.
A pesticide is a substance or mixture of substances	used to kill a chemical substance, biological agent, antimicrobial, disinfectant or device used against any pest.
Marine ecosystems	include oceans, salt marshes, estuaries, lagoons, mangroves, coral reefs, the deep sea and the sea floor.
Food webs	describe the eating relationships between living species within an ecosystem.

A mutations	is the act or process of being altered or changed.
A disease	an illness which affects a person, animal or plant. Diseases are defined as pathological conditions of a part, organ, or system of an organism resulting from various causes and characterized by specific symptoms.
Tissue	is the structural material which forms animal or plant cells.
Biochemistry	is the study of chemical processes in living organisms.
Fish hydrolysate	refers to fish carcasses.

The Prevention of Pollution by Oil

Exercise 1

Oil tankers transport some 2,000 million tonnes	of crude oil around the world by sea.
The IMO introduced measures to ensure that the majority	of oil tankers are safely built and operated.
The first oil tankers	appeared in the late 19 th century.
The first 100,000-tonne crude oil tanker	was built to cover the route from Europe to the Middle East round the Cape of Good Hope.
The potential for oil to pollute the marine environment	was officially recognised by the International Convention for the Prevention of Pollution of the Sea by Oil.
The OILPOL Convention	recognised that most oil pollution resulted from routine shipboard operations.

At the time it was normal practice	to wash the tanks out with water and then pump the resulting mixture of oil and water into the sea.
The OILPOL Convention prohibited	the dumping of oily wastes within a certain distance from land and in 'special areas'.
Although the OILPOL Convention had been ratified	pollution control still remained a minor concern.
The world was only beginning	to wake up to the environmental consequences of an industrialised society.
The Torrey Canyon ran aground	when entering the English Channel.
This resulted in the biggest oil pollution incident	ever recorded up to that time.

Exercise 2

Although accidental pollution is spectacular, it is still recognised that **operational** pollution remains the bigger threat.

Therefore a new procedure, called the **load-on-top** system, was introduced in 1969.

The **washings** resulting from tank cleaning are pumped into a special tank.

During the voyage to the **loading terminal** the oil and water separate.

The enormous growth in the transport of oil by **shipping**, but also the increasing amount of **chemicals** carried at sea, made it clear that the OILPOL Convention was no longer adequate.

The Sub-Committee on Oil Pollution became the Marine Environment Protection Committee (MEPC) and was given the same **standing** as the Maritime Safety Committee (MSC).

The Protocol of the Conference on Tanker Safety and Pollution Prevention (1978) expanded the requirements for **segregated ballast** tanks to new **crude oil** tankers.

Crude oil washing is the cleaning of cargo tanks with **high pressure** jets of crude oil.

Inert gases, low in oxygen and as a consequence incombustible, are used to replace **flammable** gases in tanks.

OPA 90 made it mandatory for all tankers **calling at** ports in the United States to have double hulls.

The IMO carried out a study into the comparative performances of the double hull and the **mid-deck** tanker designs.

Proposals were presented to the MEPC to consider an accelerated 'single-hull oil tanker **phase-out**' schedule.

The Prestige disaster also led to an extended application of the **Condition Assessment Scheme** for tankers.

The CAS requirements oblige shipowners to have documentary and **survey** procedures properly carried out and completed.

Exercise 3

It is still recognised that operational pollution remains the bigger threat.

A new procedure was called the 'Load-on-Top' system.

This system has the twofold advantage of saving oil and reducing pollution.

The washings resulting from tank cleaning are pumped into a special tank.

The water at the bottom of the tank is pumped overboard.

The most important objective is the prevention of marine pollution.

Only a limited amount of oil could enter the sea in the event of damage to the vessel.

The Protocol expanded the requirements for segregated ballast tanks to new crude oil tankers of 20,000 deadweight tonnage and above.

Segregated ballast tanks were also required where they can minimise the possibility of oil outflow.

New tankers were also required to be fitted with crude oil washing systems.

Crude oil washing is the cleaning of cargo tanks with high pressure jets of crude oil.

This system reduces the quantity of oil remaining on board after discharge.

Supplementary measures for tanker safety included the requirement for inert gas systems.

After the Exxon Valdez disaster the United States developed its Oil Pollution Act.

Some member states of IMO wanted other designs to be accepted as equivalents to double hulls.

As a result IMO carried out a study into the comparative performances of the double hull and the mid-deck tanker designs.

Proposals were presented to consider an accelerated single-hull oil tanker phase-out schedule.

The Prestige disaster also led to an extended application of the Condition Assessment Scheme for tankers.

The CAS requirements include a more transparent verification of the structural condition of the ship.

The requirements oblige shipowners to have documentary and survey procedures properly carried out.

Exercise 4

The main reasons for oil spills are shipping accidents (**groundings**, collisions, tanker incidents), bad weather (storms, hurricanes, **gusts** of wind, **seaquakes**, tropical cyclones, tornadoes, willy-willies, typhoons), intentional pollution or intentional acts of violence (**routine tank cleaning**, dumping of **oily wastes**, vandalism, **terrorism**, acts of war) and human **errors**.

The main sources of spilled oil are generally: crude oil tankers breaking, **crashing** or stranding, **product carriers** or barges wrecking, oil **wells** blowing up or oil **pipelines** breaking or leaking.

Other sources can be:

- incidents with production facilities (for instance when crude oil is changed into other products such as tar, **asphalt**, etc.)
- incidents with **oil drilling vessels** or oil platforms (**oil rigs** located in the sea or ocean pumping oil from deep in the seabed or ocean **floor**)
- incidents while **lightering** (moving oil from one vessel to another)
- natural **seepage** (when natural oil found in the ground leaks up in the water)
- **mystery spills** (people dumping oil or chemicals on purpose)

To try to prevent and help to clean up oil spills, governments **pass** laws to prevent dumping and **contamination**.

Littoral states also provide adequate reception facilities for dirty ballast and other **oily residues**.

SMCP Environmental Protection Communications

Exercise 1

I read you excellent.

I have located a vessel dumping waste.

Can you identify the polluter?

Polluter is MV Suzie.

What is course and speed of the polluter?

Course of the polluter is 012 degrees.

I will send coastguard assistance.

Stay in vicinity of the polluter.

Exercise 2

Ostend Coastguard, Ostend Coastguard, Ostend Coastguard. This is MV Marina, MV Marina, MV Marina. How do you read me? Over.

MV Marina, MV Marina, MV Marina. This is Ostend Coastguard, Ostend Coastguard, Ostend Coastguard. I read you **excellent**. Go ahead with your message. Over.

Ostend Coastguard. This is MV Marina. Located a vessel **dumping waste** in position 51 degrees 45 minutes North/003 degrees 12 minutes East at 1020 hrs UTC. Repeat 51 degrees 45 minutes North/003 degrees 12 minutes East at 1020 hrs UTC. Over.

Ostend Coastguard. This is MV Marina. Yes, I can **identify** the polluter. Polluter is MV Suzie. Over.

MV Marina. This is Ostend Coastguard. What is **course** and speed of the polluter? Over.

Ostend Coastguard. This is MV Marina. Course of the polluter is 012 **degrees**, speed 10 **knots**. Over.

MV Marina. This is Ostend Coastguard. ADVICE. Stay in vicinity of the polluter. I will send **coastguard** assistance. Over.

Ostend Coastguard. This is MV Marina. Message understood. I will **stay** in vicinity of the polluter. Out.

Exercise 3

How do you read me?

I have accidental spillage of oil.

I read you four.

Can you stop spillage?

No I cannot stop spillage.

I will send oil clearance assistance.

I will stay in vicinity of pollution.

I will cooperate with oil clearance team.

Exercise 4

Master MV Marina: Ostend Coastguard, Ostend Coastguard, Ostend Coastguard. This is MV Marina, MV Marina, MV Marina. How do you **read** me? Over.

Master MV Marina: Ostend Coastguard. This is MV Marina. I have **accidental spillage** of oil in position 51 degrees 45 minutes North/003 degrees 12 minutes East at 2325 hrs UTC extending twelve metres by three metres to North. Over.

Ostend Coastguard: MV Marina. This is Ostend Coastguard. **Say** again. Over.

Ostend Coastguard: MV Marina. This is Ostend Coastguard. What kind of **assistance** is required? Over.

Master MV Marina: Ostend Coastguard. This is MV Marina. I require **oil clearance assistance**. Over.

Ostend Coastguard: MV Marina. This is Ostend Coastguard.

ADVICE: Stay in **vicinity** of pollution and co-operate. Over.

Master MV Marina: Ostend Coastguard. This is MV Marina. Message **understood**. Out.

Reporting Pollution Incidents

Exercise 1

Ships are permitted to dispose of certain types of waste at sea but there are strict requirements.

Authorities will determine if the discharge or disposal is legal or illegal.

Pollution activities should certainly be reported.

Any dumping of plastic material into the sea is forbidden.

Garbage disposed of in the sea within twelve nautical miles from land is a pollution activity.

Exercise 2

If oil can be seen it is likely to be an illegal discharge.

If you witness any discharge from a ship involving washings of chemical or dry cargoes the pollution activity should be reported.

If you observe garbage disposed of in the sea within twelve nautical miles from the land you can report the pollution to the local port.

If you witness a vessel polluting you have to report the incident.

If possible you should provide relevant details of the incident.

If possible you should communicate when and where the incident occurred.

If possible you should note down the name of the vessel.

If you notice a vessel having accidental spillage of oil you should report type and extent of pollution.

If you observe a vessel incinerating waste you must inform the marine.

If you locate a vessel dumping chemicals you should inform the coastguard.

The Exxon Valdez Disaster: Oil Clearance Operations

Exercise 1

First dispersants were **used**.

The dispersants were **dropped** from a helicopter.

The use of dispersants was **discontinued**.

Burning was then **attempted**.

As a result of unfavourable weather conditions, no additional burning was **undertaken**.

Mechanical clean-up was **started** shortly afterwards.

The operation was **delayed**.

The skimmers were not **available** during the first 24 hours.

The decision was **made** to displace the oil with high-pressure hot water.

Microbial populations on the shoreline were **destroyed**.

Oil Clearance Operations – Methods and Devices

Exercise 1

Skimmers are designed	to recover floating oil from the surface of the water.
In situ burning is designed	to remove oil quickly and easily from the sea.
Booms are designed	to deflect oil from sensitive areas.
Dispersants are designed	to be applied by vessel or aerial spraying.
High-pressure hoses are designed	to apply hot water.

In situ burning is	an easy method to remove oil from the sea.
Booms are designed	to contain oil.
Skimmers are designed	to operate by a pump or belt system.
Dispersants are designed	to speed up shoreline clean-up.
High pressure hoses are designed	to blast oil from the surface of rocks and boulders.

Fuel Emissions

Exercise 1

Green awareness has finally reached the maritime industry. Ship companies are trying to reduce **excessive** fuel costs. The interest has arisen from financial needs because fuel is expensive.

Although it is difficult to reach a zero **emission** status, fuel **efficiency** can be obtained by several methods.

Most ships use heavy diesel oil as a combustion. Heavy fuel is in fact the **residue** obtained after distilling benzene and diesel out of the crude oil. The emission gases of heavy fuel oil are full of nitrogen (NOX) and sulphur (SOX). This causes **acid rain** and makes the hole in the **ozone layer** bigger. To reduce this form of pollution, **filters** are installed on the funnels stopping **harmful** gases from getting into the environment.

Another form of more **eco-friendly** navigation is velocity control. A **speed** reduction of 15% saves up to half of the fuel **consumption**. No wonder ship companies like this form of cost efficiency!

Electronic fuel **monitoring** is popular as well. Sometimes too much fuel is **injected** into the engine. Sometimes the mix is not right and too many harmful substances are **released** into the air. So if fuel is monitored, the engines work more efficiently and that is good for the environment!

Gas tankers, however, are the greenest means of transport at sea. They not only carry Liquid Natural Gas (LNG) to navigate but also use it as a **combustion**. The gas is loaded onto the vessel in liquid form at extremely low temperatures. During the voyage, the liquid warms up and produces **fumes**. These gases are then burnt to **propel** steam turbines.

The disadvantage is that the steam turbines are 15% less efficient than **fuel-driven** vessels. The advantage is that the methane, which produces carbon oxide after burning, is less **pollutant** than heavy fuel. It is also cost-effective because the gas which would otherwise be lost is used for propulsion.

Finally, some ship companies already use new sail power methods such as ship kites. These kites produce lots of controlled, **inexpensive** wind power. They reduce fuel consumption drastically. Although the kites do not produce **exhaust gases** and the installation is relatively cheap, wind energy is of course **dependent** on the wind.

Exercise 2

FUEL EFFICIENCY	A ship can save fuel if she monitors her speed.
GREEN AWARENESS	To be interested in the environment and to be well-informed about it
EXHAUST GAS	The waste gas from an engine, in particular that of a ship
ACID RAIN	Rain which has a lot of harmful chemicals. This rain is the result of burning substances such as coal and oil.
COMBUSTION	The process of burning
HARMFUL SUBSTANCES	Gases or materials that damage the environment
ZERO EMISSION	When a ship does not produce amounts of gas that are harmful to the environment
OZONE LAYER	A layer of air around the Earth which protects us from harmful ultraviolet sunlight

Exercise 3

to emit	emission
to mix	mixture
to reduce	reduction
to propel	propulsion
to be efficient	efficiency

to be aware	awareness
to distil	distillation
to pollute	pollutant
to install	installation
to control	control

to inject	injection
to release	release
to monitor	monitor
to load	load
to produce	production

Exercise 4

to harm	harmful
to finance	financial
to exceed	excessive
to pollute	pollutant
to navigate	navigational
to liquify / liquefy	liquid